⊙ **String** :- It is a sequence of characters.

eg :- cricket ⇒ length = 7

c	r	i	c	k	e	t
0	1	2	3	4	5	6

- $\underbrace{[A-Z]}_{26} + \underbrace{[a-z]}_{26} + \underbrace{[0-9]}_{10}$

$= 62$

- If the length of the complete string is 'n', then if you store it in array then $S.C = \Theta(n)$

char str[65]; $\Rightarrow$ $S.C = \Theta(1)$

Palindrome :-

$$\boxed{\text{Original word} = \text{Reversed word}}$$

- madam
- madam
- bat
- tab

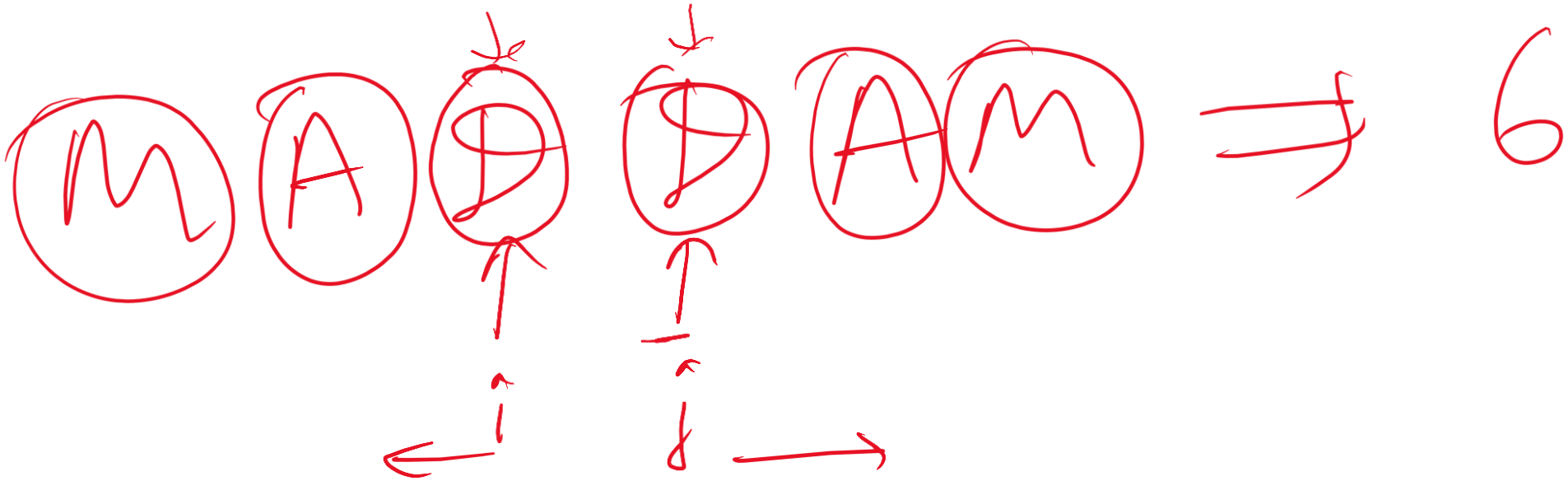
Odd Length Palindrome

M A D A M ⇒ S

i-- j++

```
while (S[i] == S[j])
{ i--;
  j++;
}
```

Even Length Palindrome :-



$\Rightarrow 6$

helper(i, i) // odd Length

helper(i, i+1) // Even Length

- **Substring** :- It is a sequence of contiguous characters.

eg :- cric ⇒ 4

Substring of Length 1 :- c, r, i, c ⇒ 4

" " " 2 :- cr, ri, ic ⇒ 3

" " " 3 :- cri, ric ⇒ 2

" " " 4 :- cric ⇒ 1

Total no of substring = $\boxed{1+2+3+4+\cdots+n} = \frac{n(n+1)}{2}$

① <u>subsequence</u> :- It is a collection of characters and those characters should be <u>in increasing order of their indices</u>.

eg :- cric ⇒ cic , rc ,

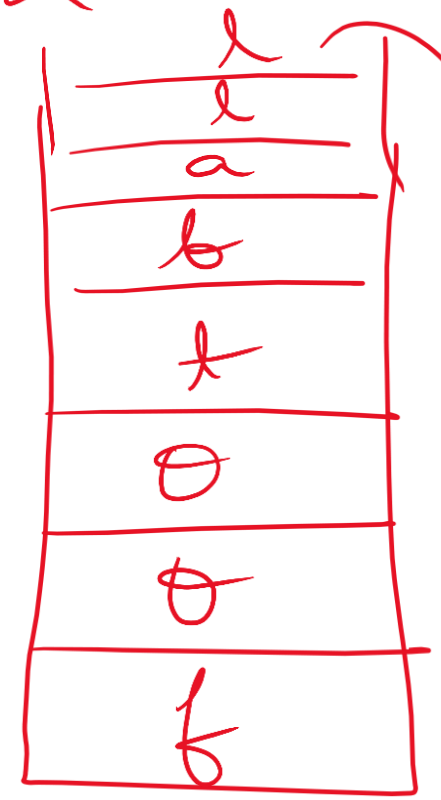
0 1 2 3   0 2 3   1 3

cc , ci , ri

0 3   0 2   1 2

① **Stack** :- It is based on LIFO (Last In First Out).

eg :- football



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